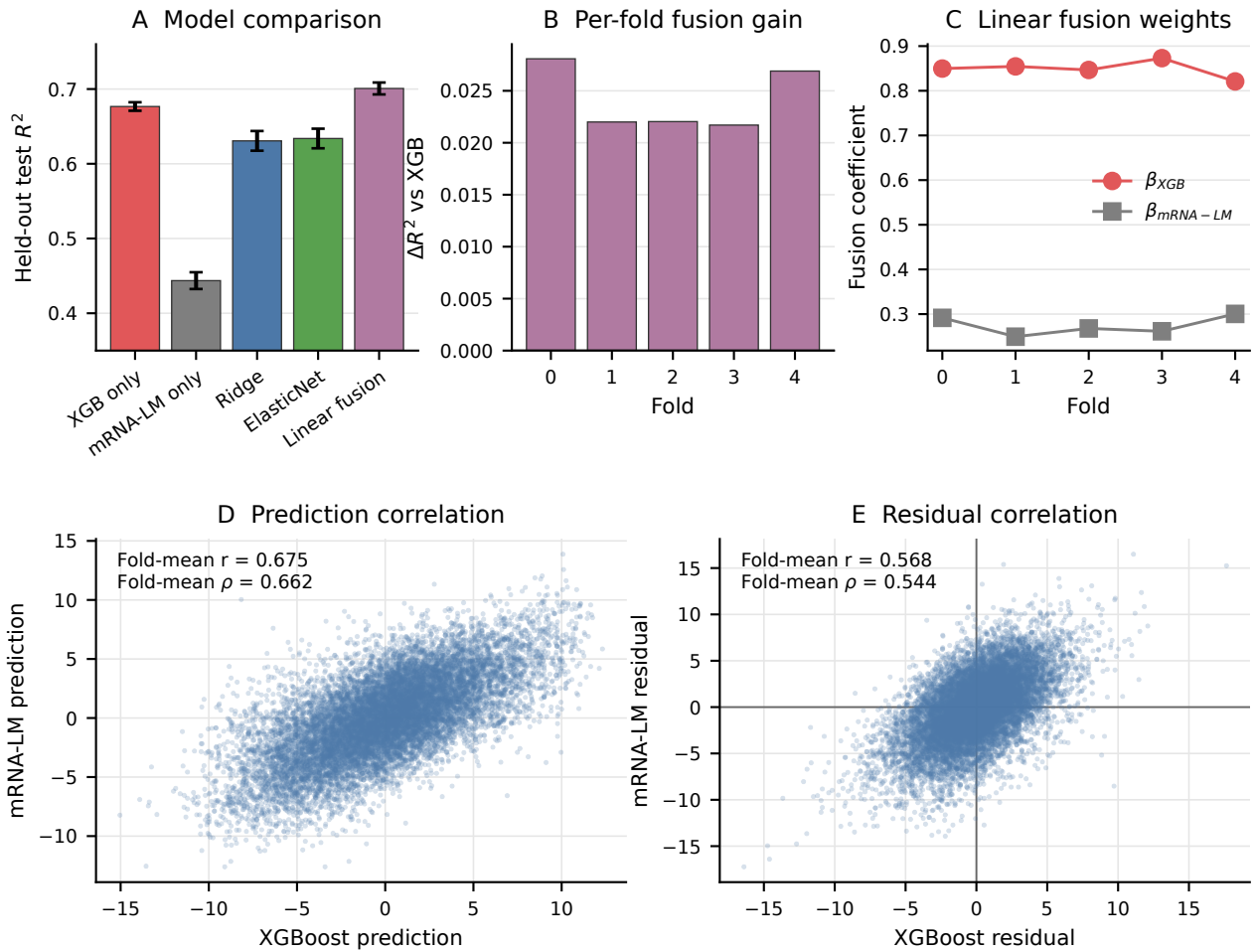

A Model comparison
Held-out test R^2
0.7
0.6
0.5
0.4
XGB only
mRNA-LM only
Ridge
ElasticNet
Linear fusion
B Per-fold fusion gain
ΔR^2 vs XGB
0.025
0.020
0.015
0.010
0.005
0.000
0
1
2
3
4
Fold
C Linear fusion weights
Fusion coefficient
0.9
0.8
0.7
0.6
0.5
0.4
0.3
β_{XGB}
$\beta_{mRNA-LM}$
0
1
2
3
4
Fold
D Prediction correlation
Fold-mean r = 0.675
Fold-mean ρ = 0.662
mRNA-LM prediction
15
10
5
0
−5
−10
−15
−15
−10
−5
0
5
10
XGBoost prediction
E Residual correlation
Fold-mean r = 0.568
Fold-mean ρ = 0.544
mRNA-LM residual
15
10
5
0
−5
−10
−15
−15
−10
−5
0
5
10
15
XGBoost residual